F78PA Probability & Statistics A

Tutorial 3 – Solutions

1. Let X denote the number of calls made until the first sale is achieved. Then $X \sim \text{Geo}\,(0.08)$, and $\mathbb{P}(X = 5) = 0.92^4 \times 0.08 \approx 0.0573$.

2. Let X be the number of free throws when the first success occurs, $X \sim \text{Geo}\,(0.7)$. So probability mass function is $p_X(x) = (0.7)(0.3)^{x-1}$ for $x = 1, 2, \ldots$

$$\begin{aligned}\mathbb{P}(X > 4) &= 1 - \mathbb{P}(X \leq 4)\\ &= 1 - [\mathbb{P}(X = 1) + \mathbb{P}(X = 2) + \mathbb{P}(X = 3) + \mathbb{P}(X = 4)]\\ &= 1 - [0.7 + 0.3(0.7) + 0.3^2(0.7) + 0.3^3(0.7)]\\ &= 1 - 0.9919\\ &= 0.0081\end{aligned}$$

3. i. $X \sim \text{Poi}(4)$, so probability mass function is $p_X(x) = \frac{4^x}{x!}e^{-4}$ for $x = 0, 1, 2, \ldots$

   ii. $\mathbb{P}(X \leq 2) = \frac{4^0}{0!}e^{-4} + \frac{4^1}{1!}e^{-4} + \frac{4^2}{2!}e^{-4} = 13e^{-4} \approx 0.2381$

   iii. $\mathbb{P}(X \geq 4) = 1 - \mathbb{P}(X \leq 3) = 1 - \frac{4^0}{0!}e^{-4} - \frac{4^1}{1!}e^{-4} - \frac{4^2}{2!}e^{-4} - \frac{4^3}{3!}e^{-4} = 1 - (71/3)e^{-4} \approx 0.5665$

   iv. $\mathbb{P}(X \geq 4 \mid X \geq 3) = \frac{\mathbb{P}(\{X \geq 4\} \cap \{X \geq 3\})}{\mathbb{P}(X \geq 3)} = \frac{\mathbb{P}(X \geq 4)}{\mathbb{P}(X \geq 3)} = \frac{1 - (71/3)e^{-4}}{1 - 13e^{-4}} \approx 0.7436$

4. i. To find k,

$$\begin{aligned}1 = \int_{x=-\infty}^{\infty} f(x)\,dx &= \int_{x=0}^{2} k(3 + 2x - x^2)\,dx\\ &= k\left[3x + x^2 - \frac{x^3}{3}\right]_{x=0}^{2} = k\left(\left(6 + 4 - \frac{8}{3}\right) - 0\right)\\ &= \frac{22k}{3}\\ \Rightarrow k &= 3/22\end{aligned}$$

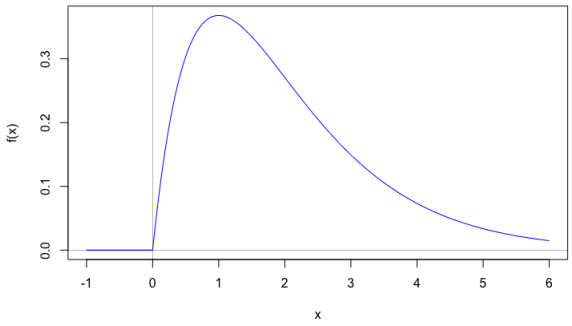
   ii.

$$\begin{aligned}\mathbb{P}(1/2 \leq X \leq 3/2) = \int_{x=1/2}^{3/2} f(x)\,dx &= \int_{x=1/2}^{3/2} \frac{3}{22}\left(3 + 2x - x^2\right)dx\\ &= \frac{3}{22}\left[3x + x^2 - \frac{x^3}{3}\right]_{x=1/2}^{3/2}\\ &= \frac{3}{22}\left(\left(\frac{9}{2} + \frac{9}{4} - \frac{9}{8}\right) - \left(\frac{3}{2} + \frac{1}{4} - \frac{1}{24}\right)\right)\\ &= \frac{3}{22}\left(\frac{45}{8} - \frac{41}{24}\right) = \frac{3}{22} \times \frac{94}{24} = \frac{47}{88} \approx 0.5341\end{aligned}$$

5.

$$f_X(x) = \begin{cases} xe^{-x} & 0 \leq x < \infty,\\ 0 & \text{otherwise.}\end{cases}$$

   (a) Sketch of f_X:



$Im\ X = \{x : f(x) > 0\} = (0, \infty)$.

(b) Integrating by parts,

$$\begin{aligned}\mathbb{P}(X \leq 3) &= \int_{x=0}^{3} xe^{-x}\,dx = [-xe^{-x}]_{x=0}^{3} + \int_{x=0}^{3} e^{-x}\,dx \\ &= -3e^{-3} + 0 + [-e^{-x}]_{x=0}^{3} = -3e^{-3} - e^{-3} + 1 \\ &= 1 - 4e^{-3} \approx 0.801\end{aligned}$$

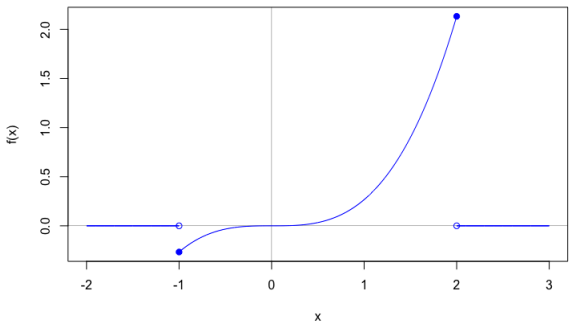
For $\mathbb{P}(X \geq 3)$, could do the integration over the range $[3, \infty)$, or simply note that since X is a continuous random variable (with a density function), $\mathbb{P}(X \geq 3) = 1 - \mathbb{P}(X \leq 3) \approx 0.199$.

For this random variable, $\mathbb{P}(X \leq 3)$ is greater than $\mathbb{P}(X \geq 3)$.

6. (a)

$$f(x) = \begin{cases} \frac{4}{15}x^3 & -1 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch:

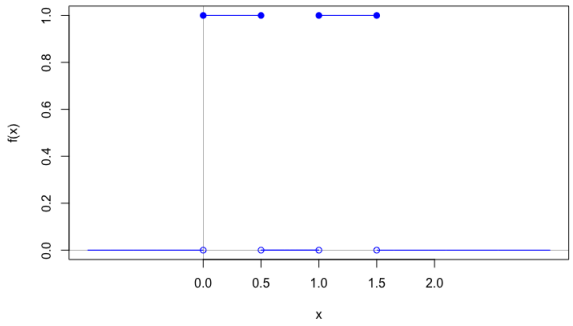


This function is **not** a density function, because $f(x) < 0$ on the interval $-1 \leq x < 0$.

(b)

$$f(x) = \begin{cases} 1 & 0 \leq x \leq \frac{1}{2}, \quad 1 \leq x \leq \frac{3}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch:



This function **is** a density function, because $f(x) \geq 0$ for all $x \in \mathbb{R}$ and $\int_{x=-\infty}^{\infty} f(x)\,dx = \int_{x=0}^{1/2} 1\,dx + \int_{x=1}^{3/2} 1\,dx = \frac{1}{2} + \frac{1}{2} = 1$.
If X is a random variable with density f, then $Im\ X = \{x : f(x) > 0\} = [0, \frac{1}{2}] \cup [1, \frac{3}{2}]$.
The cdf is, by definition, $F(x) = \mathbb{P}(X \leq x)$ for all $x \in \mathbb{R}$.
Since $Im\ X = [0, \frac{1}{2}] \cup [1, \frac{3}{2}]$, we know that:

$$\begin{aligned} \text{for } x < 0,\ F(x) &= 0, \\ \text{and for } x > \tfrac{3}{2},\ F(x) &= 1. \end{aligned}$$

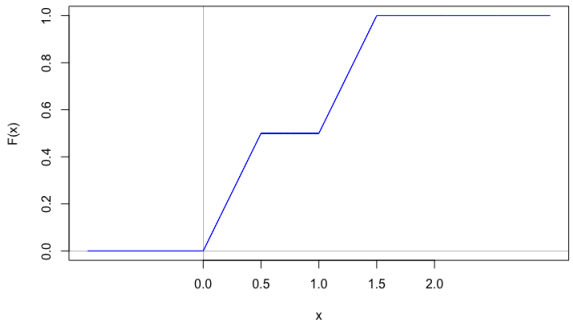
There are 3 more cases to consider:

- For $0 \leq x \leq \frac{1}{2}$, $F(x) = \int_{u=-\infty}^{x} f(u)\,du = \int_{u=0}^{x} 1\,du = x$
- For $\frac{1}{2} < x < 1$, $F(x) = \int_{u=-\infty}^{x} f(u)\,du = \int_{u=0}^{1/2} 1\,du + \int_{u=1/2}^{x} 0\,du = \frac{1}{2}$
- For $1 \leq x \leq \frac{3}{2}$, $F(x) = \int_{u=-\infty}^{x} f(u)\,du = \int_{u=0}^{1/2} 1\,du + \int_{u=1/2}^{1} 0\,du + \int_{u=1}^{x} 1\,du = \frac{1}{2} + (x-1) = x - \frac{1}{2}$

Putting this all together, we have

$$F(x) = \begin{cases} 0 & x < 0, \\ x & 0 \leq x \leq \frac{1}{2}, \\ \frac{1}{2} & \frac{1}{2} < x < 1, \\ x - \frac{1}{2} & 1 \leq x \leq \frac{3}{2}, \\ 1 & x > \frac{3}{2}. \end{cases}$$

Sketch of the cdf $F(x)$:

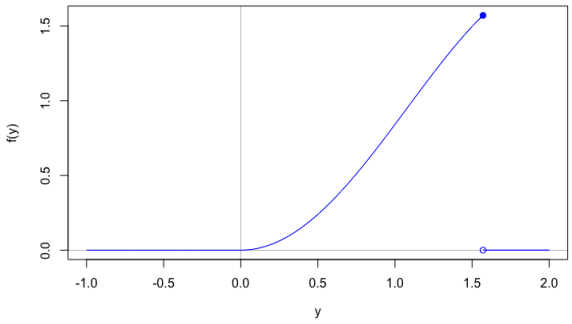


(Notice that $F(x)$ is a *continuous* function, with derivative $F'(x) = f(x)$.)

(c)

$$f(y) = \begin{cases} y \sin y & 0 \leq y \leq \frac{\pi}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch:



This function **is** a density function, because $f(y) \geq 0$ for all $y \in \mathbb{R}$ and (using integration by parts) $\int_{y=-\infty}^{\infty} f(y)\,dy = \int_{y=0}^{\pi/2} y \sin y\,dy = [-y\cos y]_{y=0}^{\pi/2} + \int_{y=0}^{\pi/2} \cos y\,dy = (-0+0) + [\sin y]_{y=0}^{\pi/2} = 0 + (1-0) = 1$.
If Y is a random variable with density f, then $Im\ Y = (0, \frac{\pi}{2}]$.
The cdf is $F(y) = \mathbb{P}(Y \leq y)$ for all $y \in \mathbb{R}$. So for $y < 0$, $F(y) = 0$, and for $y > \frac{\pi}{2}$, $F(y) = 1$.
For $0 \leq y \leq \frac{\pi}{2}$, $F(y) = \int_{t=0}^{y} t \sin t\,dt = [-t\cos t + \sin t]_{t=0}^{y} = -y\cos y + \sin y - 0$ (using integration by parts as before).
Putting it all together,

$$F(y) = \begin{cases} 0 & y < 0, \\ \sin y - y\cos y & 0 \leq y \leq \frac{\pi}{2}, \\ 1 & y > \frac{\pi}{2}. \end{cases}$$

7. Buses arrive at times $0, 15, 30, 45, 60$ (minutes past noon), and Anne arrives at X minutes after noon, where X is a random variable with distribution function

$$F_X(x) = \mathbb{P}(X \leq x) \quad = \quad \begin{cases} 0 & x < 0, \\ x^2/3600 & 0 \leq x \leq 60, \\ 1 & x > 60. \end{cases}$$

F_X is continuous (i.e. has no jumps) – you can check this by sketching the graph.

(a) Let $A = \{\text{Anne waits less than 5 minutes}\}$.
Then

$$A \quad = \quad \{X = 0\} \cup \{10 < X \leq 15\} \cup \{25 < X \leq 30\} \cup \{40 < X \leq 45\} \cup \{55 < X \leq 60\}$$

This is a union of mutually exclusive events, and so

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(X = 0) + \mathbb{P}(10 < X \leq 15) + \mathbb{P}(25 < X \leq 30) + \mathbb{P}(40 < X \leq 45) + \mathbb{P}(55 < X \leq 60) \\ &= 0 + (F_X(15) - F_X(10)) + (F_X(30) - F_X(25)) + (F_X(45) - F_X(40)) + (F_X(60) - F_X(55)) \\ &= 7/18 \end{aligned}$$

(b) Let $B = \{\text{Anne waits more than 10 minutes}\}$.
Then

$$B \quad = \quad \{0 < X < 5\} \cup \{15 < X \leq 20\} \cup \{30 < X \leq 35\} \cup \{45 < X \leq 50\}$$

This is a union of mutually exclusive events, and so

$$\begin{aligned} \mathbb{P}(B) &= \mathbb{P}(0 < X < 5) + \mathbb{P}(15 < X < 20) + \mathbb{P}(30 < X < 35) + \mathbb{P}(45 < X < 50) \\ &= \mathbb{P}(0 < X \leq 5) + \mathbb{P}(15 < X \leq 20) + \mathbb{P}(30 < X \leq 35) + \mathbb{P}(45 < X \leq 50) \\ &\quad \text{(because } F_X \text{ is continuous)} \\ &= (F_X(5) - F_X(0)) + (F_X(20) - F_X(15)) + (F_X(35) - F_X(30)) + (F_X(50) - F_X(45)) \\ &= 5/18 \end{aligned}$$

(c)

$$\begin{aligned}\mathbb{P}(\text{Anne arrives before 12:30}) = \mathbb{P}(X < 30) &= \mathbb{P}(X \leq 30) \text{ (since } F_X \text{ is continuous)} \\ &= F_X(30) = 1/4\end{aligned}$$

$$\mathbb{P}(\text{Anne arrives after 12:30}) = \mathbb{P}(30 < X \leq 60) = F_X(60) - F_X(30) = 3/4$$

Anne is more likely to arrive after 12.30 pm than before 12.30 pm.

8. (a) Suppose $X \sim \text{Exp}(\lambda)$, with $\lambda > 0$. Then X has distribution function $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ (with $F_X(x) = 0$ for $x < 0$), and to find the median m we must solve the equation $F_X(m) = 1/2$.
So we must find m such that

$$\begin{aligned} & F_X(m) = 1/2 \\ \Rightarrow \quad & 1 - e^{-\lambda m} = 1/2 \\ \Rightarrow \quad & e^{-\lambda m} = 1/2 \\ \Rightarrow \quad & -\lambda m = \ln(1/2) = -\ln 2 \\ \Rightarrow \quad & m = \frac{\ln 2}{\lambda}\end{aligned}$$

(b) Suppose X has density

$$f_X(x) = \begin{cases} 2xe^{-x^2} & x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Median m satisfies

$$\begin{aligned} F_X(m) &= 1/2 \\ \Rightarrow \quad \int_{x=-\infty}^{m} f_X(x)\,dx &= 1/2 \\ \Rightarrow \quad \int_{x=0}^{m} 2xe^{-x^2}\,dx &= 1/2\end{aligned}$$

Integrate using the substitution $u = x^2$, so $du = 2x\,dx$, and

$$\begin{aligned} F_X(m) = \int_{x=0}^{m} 2xe^{-x^2}\,ds &= \int_{u=0}^{m^2} e^{-u}\,du \\ &= \left[-e^{-u}\right]_{u=0}^{m^2} \\ &= 1 - e^{-m^2}\end{aligned}$$

So we require

$$\begin{aligned} 1 - e^{-m^2} &= 1/2 \\ \Rightarrow \quad e^{-m^2} &= 1/2 \\ \Rightarrow \quad -m^2 &= \ln(1/2) = -\ln 2 \\ \Rightarrow \quad m &= \sqrt{\ln 2} \qquad \text{(taking positive root since we know } m > 0)\end{aligned}$$

Exam Style Question

(a) f_X is a valid probability density function if $f_X(x) \geq 0$ for all x and $\int_{\mathbb{R}} f_X(x)dx = 1$. From the plot (see Figure 1) it is clear that $f_X(x) \geq 0$. It remains to show that the

integral of $f_X(x)$ is 1

$$\begin{aligned}\int_{\mathbb{R}} f_X(x)\,\mathrm{d}x &= \int_0^5 \frac{2}{75}x\,\mathrm{d}x + \int_5^{10} \frac{2}{15}\,\mathrm{d}x \\ &= \left[\frac{1}{75}x^2\right]_{x=0}^{5} + \frac{2}{15}(10-5) \\ &= \frac{1}{75}5^2 + \frac{2}{15}(5) = \frac{1}{3} + \frac{2}{3} = 1\end{aligned}$$

As required. [1 mark]

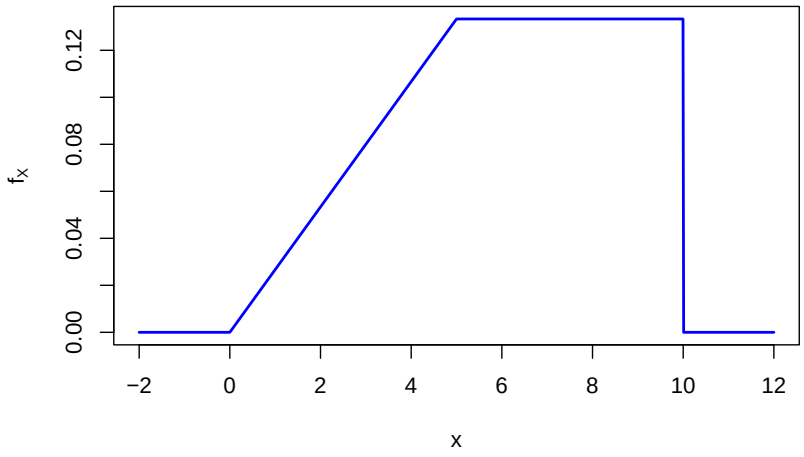


Figure 1: Plot of the probability density function $f_X(x)$

(b) The image of X is $\mathrm{Im}(X) = [0, 10]$ (this is the set of values where $f_X(x) \neq 0$).
So for $x < 0$, $F_X(x) = 0$ and for $x > 10$, $F_X(x) = 1$.
There are two more cases to consider:
For $0 \leq x \leq 5$:

$$F_X(x) = \int_0^x \frac{2}{75}u\,\mathrm{d}u = \left[\frac{1}{75}u^2\right]_{u=0}^{x} = \frac{x^2}{75}.$$

For $5 < x \leq 10$:

$$F_X(x) = \frac{5^2}{75} + \int_5^x \frac{2}{15}\,\mathrm{d}u = \frac{1}{3} + \frac{2}{15}(x-5) = \frac{2}{15}x - \frac{1}{3}.$$

Therefore

$$F_X(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{75}x^2, & 0 \leq x \leq 5, \\ \frac{2}{15}x - \frac{1}{3}, & 5 < x \leq 10, \\ 1, & 10 < x. \end{cases}$$

[3 marks]

(c) i)

$$\mathbb{P}(X > 4) = 1 - F_X(4) = 1 - \frac{1}{75}4^2 = \frac{59}{75}.$$

[1 mark]

ii) The median occurs when $F_X(m) = 0.5$ and since $F_X(5) = \frac{1}{3}$ we must have that $m > 5$. Therefore, we need to solve

$$F_X(m) = \frac{2}{15}m - \frac{1}{3} = 0.5.$$

Solving this equation gives $m = 6.25$. [1 mark]

(d) Let Y be the number of batteries that stop working within 40 hours in the said electronic device. Using the result from part b(i) above, $Y \sim \text{Bin}(2, \frac{16}{75})$.

Therefore

$$\begin{aligned}
\mathbb{P}(\text{device stops working within 40 hours}) &= \mathbb{P}(Y \geq 1) \\
&= 1 - \mathbb{P}(Y = 0) \\
&= 1 - \binom{2}{0}\left(\frac{16}{75}\right)^0\left(\frac{59}{75}\right)^2 \\
&= \frac{2144}{5625}.
\end{aligned}$$

[2 marks]

[Total 8 marks]